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Screening with Persuasion

Contents

1 Big picture

- **Question.** If a seller can choose both a menu of products and the information buyers receive about their own willingness to pay, what does the optimal selling mechanism look like?
- **Setting.** The paper studies a canonical nonlinear pricing environment with vertical differentiation. Buyers value quality q with utility $vq - p$, where v is initially private to the seller only through a latent distribution. The seller controls both the public menu and the information structure.
- **Core challenge.** In standard screening models, a seller uses a rich menu of qualities and prices to sort buyers. Here, however, a richer menu forces the seller to reveal more information to buyers, which raises informational rents. The seller now trades off allocative efficiency against information-rent compression.
- **Main conceptual contribution.** The paper integrates **screening** and **Bayesian persuasion** in one mechanism-design problem. Instead of treating buyer information as exogenous, the seller endogenously coarsens it.
- **Main theorem 1.** The optimal mechanism always has a **finite menu**, even when the underlying value distribution is continuous. The seller never wants to screen on an open interval of values.
- **Main theorem 2.** Pooling is not only a continuum phenomenon. Even with a finite value distribution, pooling must be optimal whenever entropy exceeds a threshold $E^* \in (\log_2 8, \log_2 9)$, so complete disclosure can survive only for sufficiently coarse distributions.
- **Main theorem 3.** Under a separable cost function, $c'''(q) \geq 0$, and a **modest-tails** condition on the value density, the optimal menu is a **single-item menu**. In signal space, the optimal information policy becomes binary: buy or do not buy.
- **Additional structural result.** Any optimal discrete menu is **convex**: quality increments become larger as one moves up the menu.
- **Main economic intuition.** Pooling slightly distorts allocation only at second order, but it reduces informational rents at first order. That local asymmetry makes coarse menus profitable.
- **Practical interpretation.** In digital markets, recommendation systems and common posted menus may optimally work together. Sellers often benefit from giving buyers only coarse information and offering only a small number of products.
- **Economic takeaway.** The classic continuous-menu logic of Mussa and Rosen is fragile once the seller controls what buyers learn about themselves. Information compression and product-line compression become complements.

2 Model environment and key objects

2.1 Environment and payoffs

A seller offers qualities $q \in [q_\ell, q_h]$ to a continuum of buyers. Buyer's utility from quality q at price p is

$$u(v, q, p) = vq - p. \tag{1}$$

Buyer values $v \in [v_\ell, v_h]$ are distributed according to a common prior F, which the paper calls the **latent distribution**.

Interpretation. In the standard nonlinear-pricing model, buyers know their own v. Here they initially know only the prior. The seller controls how much of their value information they learn before choosing from the menu.

2.2 Cost of quality

The seller produces a distribution of qualities G at cost

$$C(G) : \Delta[q_\ell, q_h] \to \mathbb{R}_+. \tag{2}$$

A benchmark separable specification is

$$C(G) = \int c(q)\, dG(q), \tag{3}$$

with $c(\cdot)$ convex. More generally, the paper assumes $C(G)$ is increasing in the **increasing convex order**: higher or more dispersed quality distributions are more costly.

Interpretation. The main finite-menu result does not rely on a very special technology. It covers both variable-inventory models and fixed-inventory models.

2.3 The mechanism: menu plus information

The selling mechanism is (P, Q, S):

- Q: the set of qualities offered;
- $P(q)$: the public price of quality q, with $P(0) = 0$;
- $S(v)$: an information structure mapping latent values into signal distributions.

Given a signal s, the buyer's posterior expected value is

$$\bar{v}(s) = \mathbb{E}[v \mid s]. \tag{4}$$

Interpretation. The seller cannot use personalized prices contingent on the realized signal. Information is used only to steer buyers through a common menu.

2.4 Buyer choice and induced distributions

A buyer with posterior mean $\bar{v}$ chooses the quality that maximizes utility:

$$q(\bar{v}) \in \arg\max_{q \in Q} \{\bar{v}q - P(q)\}. \tag{5}$$

The information structure induces a distribution of posterior means, denoted $\bar{F}$, and therefore an induced distribution of sold qualities, denoted $\bar{G}$. Seller profit is

$$\mathbb{E}[P(q(\bar{v}))] - C(\bar{G}). \tag{6}$$

Interpretation. The seller chooses both how buyers are bunched in posterior-value space and how those posterior values map into product choices.

2.5 Direct-mechanism representation

The paper translates the menu problem into a direct allocation $(q(\bar{v}), p(\bar{v}))$. Incentive compatibility implies monotonicity of the allocation and the envelope formula

$$p(\bar{v}) = \bar{v}\, q(\bar{v}) - \int_{v_\ell}^{\bar{v}} q(s)\, ds. \tag{7}$$

Interpretation. As in standard mechanism design, once the allocation rule is monotone, payments are pinned down by the envelope condition. The novelty is that the seller also chooses the distribution $\bar{F}$ itself.

2.6 Feasible information and the optimization over two distributions

By Blackwell's theorem, a distribution of posterior means $\bar{F}$ is feasible if and only if it is a **mean-preserving contraction** of the latent distribution F:

$$F \prec \bar{F}. \tag{8}$$

The seller can therefore rewrite the problem as a joint optimization over the distribution of posterior values $\bar{F}$ and the distribution of supplied qualities $\bar{G}$:

$$\max_{\bar{F}, \bar{G}} \int_0^1 \bar{F}^{-1}(t)(1-t)\, d\bar{G}^{-1}(t) + \bar{G}^{-1}(0)\bar{F}^{-1}(0) - C(\bar{G}), \tag{9}$$

subject to $F \prec \bar{F}$ and implementability.

Interpretation. This is the paper's key reformulation. Revenue becomes bilinear in the inverse distributions of posterior values and qualities. Once written this way, local pooling arguments become transparent.

2.7 Interpretation in the digital economy

The paper links the model to three digital-market features:

- sellers often know more than buyers about product-match quality;
- recommendation systems act as information structures;
- many platforms use common menus rather than personalized prices.

Interpretation. Think of a platform that recommends one of a few subscription tiers or product bundles to users, while still posting the same public menu to everyone.

3 Model and equilibrium (all equations + interpretation)

3.1 A first finite-value example: why pooling can beat disclosure

The paper first studies a finite model with values $\{v_1, \ldots, v_N\}$ and qualities $\{q_1, \ldots, q_N\}$. The discrete virtual value is

$$\phi_i = v_i - (v_{i+1} - v_i)\frac{\sum_{j=i+1}^{N} f_j}{f_i}. \tag{10}$$

Without persuasion, efficient screening is optimal when virtual values are increasing and positive. Prices under complete disclosure are

$$P(q_i) = v_1 q_1 + \sum_{j=2}^{i}(q_j - q_{j-1})v_j. \tag{11}$$

The paper then considers **lower pooling**: pool the two lowest values into one posterior mean. For uniformly spaced values, pooling beats complete disclosure if

$$f_2 < \sqrt{f_1} - f_1, \tag{12}$$

and if the values are also uniformly distributed, this already holds for $N \geq 5$.

Interpretation. Pooling adjacent values reduces information rents paid to all higher-value buyers. The gain from rent reduction can dominate the local efficiency loss from bunching, especially when the mass points are small or the value grid is fine.

3.2 Theorem 1: the optimal mechanism is finite

Theorem 1. Every optimal mechanism is finite.

The proof intuition is the core insight of the paper. Suppose the mechanism screens over a small interval of values. Pooling that interval has two effects:

- it distorts allocation only locally, which generates a **second-order** loss in virtual surplus;
- it reduces informational rents on all higher values, which generates a **first-order** gain.

Therefore, screening on an open interval can never be optimal.

Interpretation. Classical screening says "separate types except where ironing forces pooling." This paper reverses that perspective: once information is endogenous, local separation is too expensive because it forces the seller to reveal too much.

3.3 Proposition 3: optimal mechanisms are discrete

Before proving finiteness, the paper proves a weaker statement.

Proposition 3. Every optimal mechanism is discrete.

The argument is a local deviation. If posterior means vary continuously on an interval, the seller can pool a tiny interval of values and qualities. The revenue loss from smoothing the allocation is negligible, but the rent reduction is strictly beneficial.

Interpretation. The discrete result is already powerful: even with a continuum of underlying values, optimal expected values jump across finitely or countably many pooled bins.

3.4 Proposition 4: optimal menus are convex

Write the discrete menu as posterior values $\bar{v}_k$ and corresponding qualities $\bar{q}_k$. The menu is **convex** if

$$\bar{q}_{k+1} - \bar{q}_k > \bar{q}_k - \bar{q}_{k-1}, \qquad \bar{q}_0 = 0. \tag{13}$$

Proposition 4. Every optimal mechanism generates a convex menu.

Interpretation. Quality jumps get larger as one moves up the ladder. This prevents accumulation points at the top and helps upgrade discreteness into full finiteness. Economically, if adjacent qualities were too close together high in the menu, pooling them would again save too much in information rents relative to the allocative cost.

With fixed inventory on $[q_\ell, q_h]$, the paper further obtains

$$K < \frac{q_h}{q_\ell}, \tag{14}$$

which bounds the number of menu items.

Interpretation. In a fixed-inventory environment, convexity gives a concrete finite upper bound on menu length.

3.5 Corollary 3: optimal information is a monotone partition

The paper next shows that there exists an optimal mechanism in which the value space is partitioned into intervals, and each interval is either:

- fully disclosed, or
- pooled into a single posterior mean.

Formally, if $[t_i, t_{i+1})$ is a pooled interval in quantile space, then the posterior mean is

$$\bar{v}_i = \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} F^{-1}(t)\, dt. \tag{15}$$

Interpretation. The optimal signal is not an arbitrary garbling. It can be represented as telling the buyer only which value interval they belong to. The paper stresses, however, that this monotone-partition result is *not* what drives theorem 1.

3.6 Theorem 2: entropy forces pooling

Let F be a finite distribution on an equally spaced grid, with entropy

$$\mathcal{E}(F) = -\sum_{k} f_k \log_2 f_k. \tag{16}$$

Theorem 2. If the entropy of F is larger than $E^* \in (\log_2 8, \log_2 9)$, then the optimal mechanism must pool values.

The proof works through hazard rates. A distribution that barely keeps complete disclosure optimal must satisfy a recursive lower bound on adjacent hazard rates, and those constraints sharply limit how much entropy such a distribution can have.

Interpretation. Complete disclosure can survive only when the distribution is very coarse. A distribution with variation roughly comparable to a uniform distribution over nine values already forces some pooling.

3.7 Theorem 3: when the optimal menu has a single item

For sharper characterization, the paper specializes to separable cost and imposes a **modest-tails** condition on the density:

$$f'(v) < 0 \;\Rightarrow\; f''(v) \le 0. \tag{17}$$

Theorem 3. If the value distribution has modest tails and $c'''(q) \ge 0$, then the optimal mechanism is a single-item menu.

Interpretation. Two forces make extensive pooling optimal:

- if marginal cost is convex enough, efficient qualities are compressed together;
- if the value distribution does not have heavy upper tails, there is not enough mass far away from the mean to justify carving out many separate menu items.

In signal space, theorem 3 says the seller optimally gives buyers a **binary recommendation**: buy the one offered item or do not buy.

3.8 The pooled-allocation intuition behind theorem 3

If all values are pooled, let $q_m = c'^{-1}(m_v)$ be the quality corresponding to the mean value, and let $q_h = c'^{-1}(v_h)$ be the efficient quality for the highest type. Separating a small interval of high values creates informational rents

$$U = q_m(v_h - m_v) \tag{18}$$

but also gains in total surplus

$$\Delta S = v_h(q_h - q_m) - \big(c(q_h) - c(q_m)\big). \tag{19}$$

Pooling remains attractive when U dominates ΔS only weakly, which is exactly what happens when the upper tail is modest and the marginal cost is sufficiently convex.

Interpretation. A single-item menu is not saying all buyers are the same. It says that, after accounting for information rents, the seller often prefers to communicate only a coarse buy/no-buy signal.

4 What makes the results credible (assumptions, robustness, and limits)

4.1 Why the finite-menu result is not a special-cost artifact

One of the strongest features of the paper is that theorem 1 is *not* driven by engineering a particular production technology. The finite-menu logic holds for a broad class of cost functions that are increasing in the increasing convex order, including:

- standard separable variable-inventory costs;
- fixed-inventory environments in which the seller can only reallocate and discard existing goods.

Interpretation. The paper is not saying "finite menus arise because production has integer frictions." It is saying "finite menus arise because revealing information is costly in rent terms."

4.2 Why theorem 1 does not rely on the monotone-partition result

The monotone-partition result is elegant, but the authors are explicit that the main finite-menu theorem does not need it. Theorem 1 is driven by:

- the local pooling argument for discreteness;
- the convex-menu result that eliminates accumulation points.

Interpretation. This matters because it shows the paper's central logic is robust to the exact shape of optimal signal partitions.

4.3 Key assumptions doing the heavy lifting

Several assumptions are central:

- **Public menu / no personalized pricing.** The seller can personalize information, but not posted prices.
- **Vertical differentiation.** Utility is $vq - p$, so quality is a common ranking dimension.
- **Bounded supports.** Bounded quality and value supports help convert discreteness into finiteness.

- **Seller control of information.** The seller chooses the signal structure instead of taking it as exogenous.

Interpretation. If the seller were allowed third-degree price discrimination after observing signals, the problem would be very different. If preferences were horizontally differentiated rather than vertical, the information policy would likely move in the opposite direction.

4.4 What the paper can and cannot fully pin down

The paper establishes *existence and structure* results:

- the optimal menu is finite;
- pooling is often strictly optimal;
- convex menus and, under conditions, single-item menus emerge.

But without additional assumptions, it does *not* fully characterize:

- the exact pooled intervals for every environment;
- the exact number of items for a general cost function;
- the full comparative statics for every possible value distribution.

Interpretation. The paper is strongest on the shape of the solution set and the economic force behind pooling, rather than a closed-form menu in every case.

4.5 Relation to the classic screening literature

Relative to Mussa and Rosen, the striking shift is:

- **without persuasion:** irregularity leads to bunching, but otherwise menus tend to be fine;
- **with persuasion:** pooling becomes a generic profit-improving tool because it saves information rents.

The paper also relates to Wilson's result that coarse menus generate only second-order efficiency losses. Here the new point is that rent savings are first order, so coarse menus are not just approximately efficient – they are often strictly profit maximizing.

Interpretation. The paper gives a new rationale for why firms in information-rich markets may deliberately offer short product lines.

4.6 Important limitation highlighted by the authors

In the conclusion, the authors emphasize that the results are tied to **vertical differentiation**. Under pure horizontal differentiation, for example

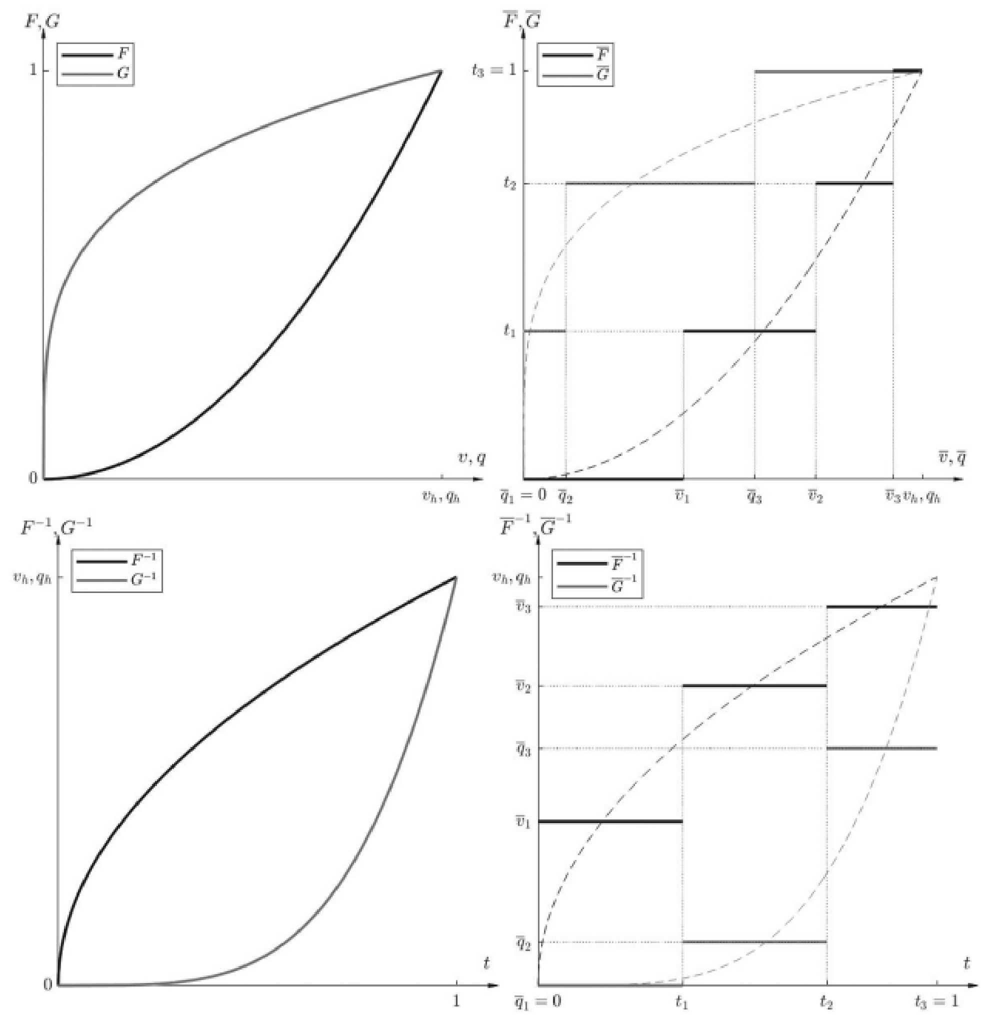
$$u(v, q) = u - (v - q)^2,$$

the seller would typically want to reveal values fully and match buyers to their ideal variety.

Interpretation. The paper is best read as a theory of coarse menus and coarse recommendations for vertically ranked products, not as a universal theory of all platform recommendations.

5 Figures and key theoretical results

5.1 Figure 1: fixed inventory example



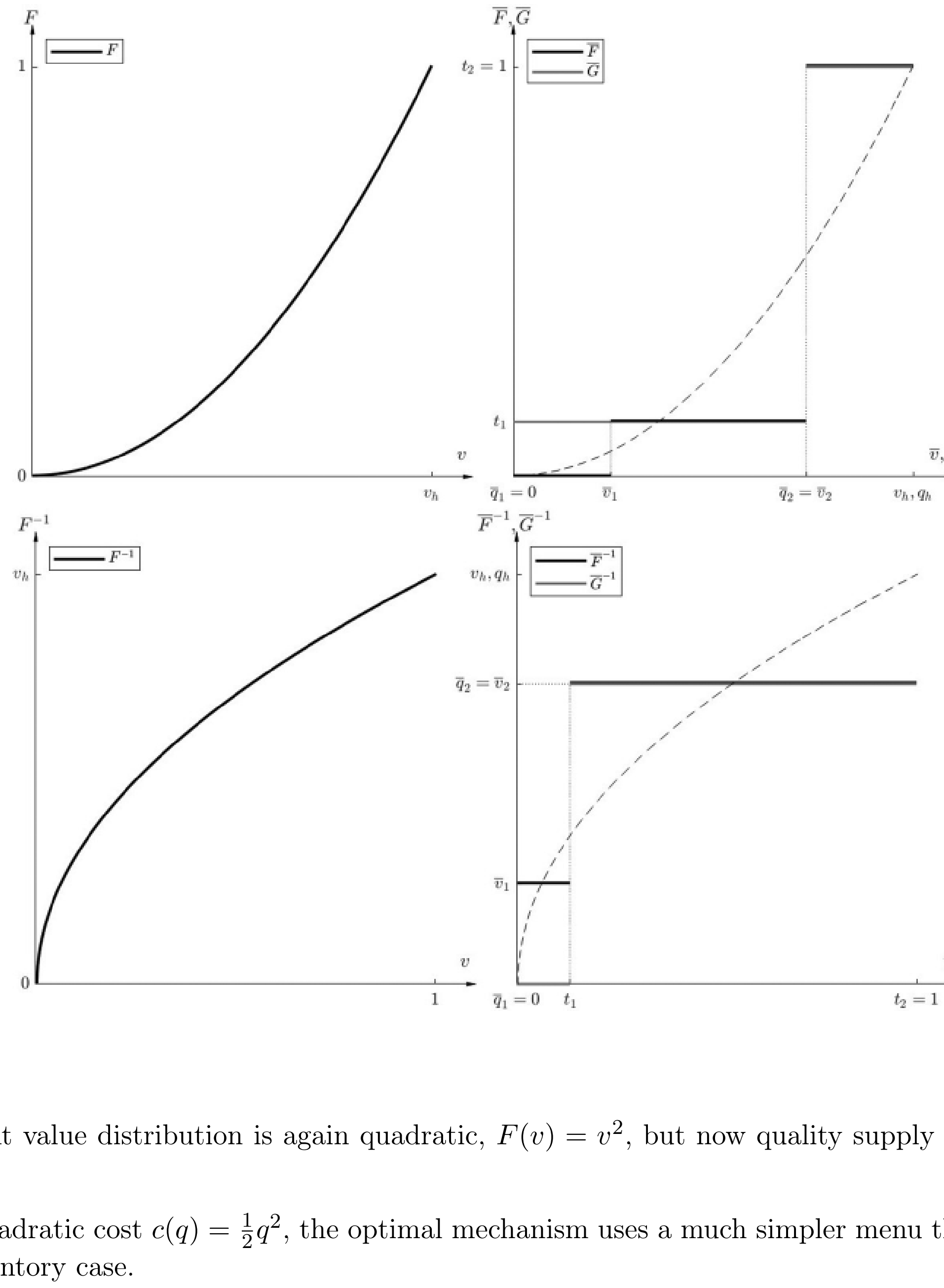
Read:

- The top-left panel shows the exogenous distributions of values $F(v) = v^2$ and qualities $G(q) = q^{1/4}$.
- The top-right panel shows the optimal pooled distributions $\bar{F}$ and $\bar{G}$.
- The bottom panels translate the same idea into quantile space. The optimal distributions jump

at the same quantiles.

Economic message. With fixed inventory, the seller cannot redesign the stock of goods, but can redesign how qualities are pooled and how much buyers learn. The figure makes the monotone-partition and finite-menu logic visually transparent.

5.2 Figure 2: variable supply and the single-item intuition



Read:

- The latent value distribution is again quadratic, $F(v) = v^2$, but now quality supply is endogenous.
- Under quadratic cost $c(q) = \frac{1}{2}q^2$, the optimal mechanism uses a much simpler menu than in the fixed-inventory case.
- The right-hand panels show that the optimal distributions collapse to essentially one positive quality plus exclusion.

Economic message. Once the seller also chooses the quality supply, convex production costs strengthen the incentive to compress the menu. This figure gives a concrete picture of the logic behind theorem 3.

5.3 Figure 3: hazard rates and the entropy bound

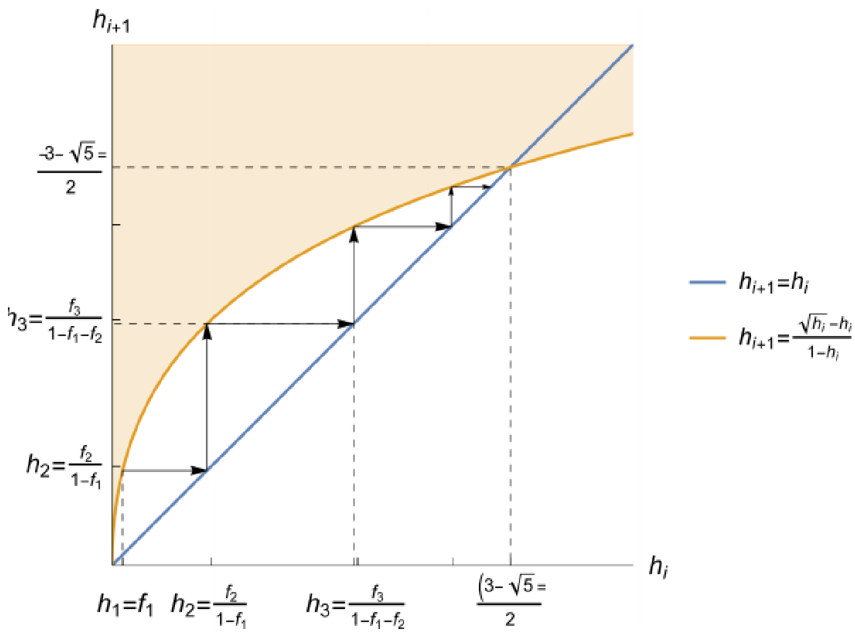


FIG. 3.—Behavior of hazard rate h_i for critical distribution $\hat{f}$.

Read:

- The blue 45-degree line is $h_{i+1} = h_i$.
- The orange curve is the binding lower-bound relation for hazard rates under complete disclosure.
- Starting from a small hazard rate, the recursion forces hazard rates to rise quickly toward a fixed point.

Economic message. The figure explains why complete disclosure can only survive for low-entropy distributions. A distribution that keeps disclosure optimal cannot spread probability mass too finely across many states.

5.4 Key theorem map

Result	What it says economically
Theorem 1	Endogenous information compression makes the optimal menu finite.
Proposition 4	Menu increments must be convex, so product lines spread out as quality rises.
Theorem 2	High-entropy value distributions cannot be served with complete disclosure.
Theorem 3	Under modest tails and convex marginal cost, a single-item menu is optimal.

Economic message. The paper moves from a local pooling argument to a global characterization: finite menu $\rightarrow$ convex menu $\rightarrow$ entropy bound $\rightarrow$ single-item condition.

6 Short summary

- The seller jointly chooses a public menu and what buyers learn about their own value.
- Because richer menus require richer information, product variety and information disclosure become complements.
- Pooling values reduces informational rents at first order, while the allocative distortion is only second order locally.
- Therefore the optimal mechanism is finite, often pooled, and under strong enough conditions collapses to one item plus exclusion.
- The paper gives a theory for why information-rich sellers may rationally offer short menus and coarse recommendations.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that once the seller controls buyer information, the canonical screening problem changes fundamentally. Optimal mechanisms no longer generically resemble continuous nonlinear-pricing schedules. Instead, they optimally compress information and compress product variety at the same time.

7.2 Economic intuition

A fine menu is expensive because it forces the seller to reveal too much about the buyer's type. Pooling a few nearby types together sacrifices little allocative efficiency, but it sharply cuts the rents the seller must leave to all higher types. That logic pushes the mechanism toward few signals and few products.

7.3 Takeaway

Screening with Persuasion shows that menu design and information design should be analyzed jointly. In vertically differentiated markets with recommendation systems and posted menus, the optimal policy is often not to help buyers identify their exact best product, but to tell them only enough to support a coarse, profit-maximizing choice.